Baseline Retrieval vs. Cross-Encoder Re-Ranking: A Two-Stage RAG Study on the Gloomhaven Rulebook

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1 Setup

Corpus	Gloomhaven rules reference, 74,916 characters
Sectioning	39 headed sections detected from the layout
Chunking	550 characters, section-aware; 167 chunks
Chunk sizes	min 160 / median 490 / max 571 characters
Bi-encoder	`all-MiniLM-L6-v2` (384-dim, local)
Cross-encoder	`cross-encoder/ms-marco-MiniLM-L-6-v2`
Generator	`gemini-3.8-flash`
Vector store	normalised matrix, cosine via one NumPy matmul

Both pipelines share stage-1 retrieval and call the same generation function with an identical prompt string, so the only variable between Answer 1 and Answer 2 is which three chunks are selected.

A rulebook was chosen over a policy or legal document deliberately. The failure modes this assignment asks for – keyword traps, exceptions, negation – all require a document that defines many near-synonymous terms precisely and then qualifies them. Gloomhaven separates *exhausted* from *eliminated* and *lost* from *discarded*, and states most rules as a general case followed by a restriction located paragraphs away. A regulatory PDF has similar structure but spreads it over far more text, meaning a larger corpus for the same number of usable test cases.

2 Architecture Comparison

A bi-encoder passes each text through the transformer separately and mean-pools the token states into one fixed vector. Chunk vectors depend on nothing but the chunk, so they can be computed once at index time; a query is embedded at run time and compared by cosine similarity. The entire interaction between query and chunk is a single dot product between two vectors produced in ignorance of each other. Whatever mean-pooling discarded is unavailable at scoring time, and pooling is lossy in a specific way: it is order-insensitive and dominated by tokens carrying the most semantic weight. Function words that invert meaning – *not*, *cannot*, *except* – contribute little relative to content words, which is why "a stunned figure may use items" and "a stunned figure cannot use items" land close together in the space.

A cross-encoder produces no comparable vectors at all. It concatenates query and chunk into one sequence and runs full self-attention over the pair, so every query token attends to every chunk token. The output is a single relevance score for that specific pairing. This lets the model condition on exact token alignment: *cannot* in the chunk can attend directly to *may* in the query and change the score, which is impossible once both have been pooled.

The cost of that expressiveness is that nothing can be precomputed. A bi-encoder index is built once; a cross-encoder score exists only after the query arrives and must be recomputed for every candidate. This asymmetry is the entire justification for the two-stage design, and Section 6 quantifies it.

3 Query Set and Method

Ten queries were written against rules located by reading the document first, then phrased deliberately in everyday synonyms rather than the rulebook's own vocabulary ("strike back" for *retaliate*, "shut entryway" for *closed door*). A gold chunk was identified by hand for each query, and a query counts as answered only when the gold chunk appears in the top 3.

Phrasing matters more than it first appears. If queries reuse the document's own vocabulary, lexical overlap alone ranks the correct chunk first and both pipelines succeed, measuring nothing. Writing "strike back" instead of *retaliate* and "shut entryway" instead of *closed door* removes that shortcut and forces retrieval to work semantically – the condition under which the architectures actually differ. Queries were written after reading the rulebook and locating a genuinely tricky rule, rather than by guessing and hoping some would fail.

4 Qualitative and Rank-Shift Analysis

4.1 Outcome across the query set

Verdict	bi → ce	Query
Fixed by re-rank	7 → 1	am I allowed to skip part of what my card tells me to do?
Fixed by re-rank	6 → 2	may a stunned figure drink a healing potion on its turn?
Fixed by re-rank	7 → 3	what occurs if a monster has nothing it is able to aim at?
Artifact (see 4.4)	4 → 3	may I stockpile two of the same piece of equipment?
Both correct	1 → 1	does mending someone's injuries also cure the venom?
Both correct	2 → 1	can a beast walk through a shut entryway?
Broken by re-rank	2 → 6	do I get a bigger hand of cards as I grow more powerful?
Broken by re-rank	2 → 4	do sealed entrances open when someone steps onto them?
Stage-1 miss	—	if I strike back at someone who hit me, does that count as attacking?
Stage-1 miss	—	is there a ceiling on restoring a wounded hero's vitality?

Three genuine rescues, two regressions, two failures that re-ranking could not reach.

4.2 Rank shifts on the rescued queries

Query	Gold chunk	bi-rank	ce-rank	bi-score	ce-score
skip part of an action	17	7	1	0.330	−3.475
stunned figure, potion	43	6	2	0.344	−1.680
monster with no target	71	7	3	0.476	−1.628

Note the bi-encoder scores: on the first query the gold chunk scored 0.330 against 0.396 for the top-ranked distractor – a gap of 0.066 across seven rank positions.

This is a predictable consequence of mean-pooled cosine similarity. Pooling maps any passage about taking a turn into approximately the same region of the space, so chunks sharing a topic share a score, and the metric has no resolution left to distinguish the chunk that answers the question from six that merely discuss it.

Cross-encoder scores on the same candidates span −3.5 to −11.3 – several points rather than hundredths. Scoring each pair jointly, the model is not committed to placing all turn-related chunks together, and separates them decisively.

4.3 Answer 1 vs Answer 2

On the stun query the baseline did not hallucinate – it correctly reported that it could not answer:

> **Answer 1 (baseline):** "the answer is not contained within the text… the context does not mention the 'stunned' condition or rules regarding drinking potions."
>
> **Answer 2 (re-ranked):** "No, a stunned figure cannot drink a healing potion on its turn. *STUN: The figure cannot perform any abilities or use items except, if a character, to perform a long rest.*" (chunk 43)

That behaviour is a product of the prompt, which instructs the model to state explicitly when context is insufficient rather than guess. Under that constraint, retrieval failure degrades into abstention instead of fabrication – the safer outcome, since a user told "this is not in the context" knows to look elsewhere, while a user given a confident wrong answer does not.

It would be wrong to conclude the prompt solves the problem. The same three chunks under a less careful prompt, or with a model more disposed to fill gaps, would plausibly have produced an answer assembled from the general action rules – fluent, specific and wrong. And abstention is still failure: the user asked an answerable question about a rule present in the document and received nothing. The prompt converted a wrong answer into a missing one. Retrieval quality sets a ceiling on answer quality that prompt engineering can protect but cannot raise.

The two regressions sharpen this. Both involve a general rule and a specific instance of it. For the hand-size query the cross-encoder promoted chunks about levelling up and card availability over the chunk stating that hand size is fixed at the value printed on the character mat; for the pressure-plate query it preferred passages about doors and room reveals over the specific rule on sealed entrances. In both cases the demoted chunk answers the question tersely, in a single clause, while the promoted chunks discuss the topic at greater length. A re-ranker trained on web passage relevance appears to reward elaboration on the query's subject, which is usually a sound heuristic and is wrong exactly when the answer is a one-line exception. The failures are not random: they are the same mechanism that produces the rescues, applied where it does not fit.

4.4 Failure categories covered

Spec category	Query	Evidence
Keyword trap / distractor	am I allowed to skip part of what my card tells me to do?	Baseline top 3 (chunks 18, 11, 16) are saturated with *card*, *action*, *turn*; the gold chunk 17 phrases the rule as "free to not perform any part". Rank 7 → 1.
Exceptions & corner cases	[not yet isolated – see note below]	—
Negation & constraints	may a stunned figure drink a healing potion on its turn?	Gold rule is stated as *cannot... except*; baseline abstains, re-rank quotes it. Rank 6 → 2.
Buried chunk (ranks 4–10 promoted)	what occurs if a monster has nothing it is able to aim at?	Gold chunk 71 at rank 7, promoted to 3. Also observed at ranks 6, 7 and 4 on the other queries.

TODO: the exceptions/corner-case category currently shares the stun query with negation. Add a dedicated query where the baseline returns a general rule and the governing "except when..." sits in a different chunk, then fill the row above.

4.5 A measurement artifact worth reporting

The duplicate-items query appears in the table as fixed, but it is not. The rulebook states the rule twice in different words:

> "You cannot own more than 1 copy of any item card."
> "No character may own duplicates of the same item."

The gold substring matched only the first. The baseline's top 3 contained the second, so its answer was correct while being scored wrong.

Any evaluation testing for a fixed substring inherits this problem, and the bias is not symmetric. A document stating a rule once in the gold string's words is scored correctly; a document that paraphrases itself produces false failures for whichever pipeline retrieved the paraphrase. Since the measurement was built around cases where the baseline was expected to fail, errors of this kind flatter the re-ranker. Two fixes exist: accept a set of acceptable phrasings per query, or abandon chunk identity as the criterion and judge the generated answer against the rule itself. The second is more work and more faithful, since retrieving a different chunk supporting the same correct answer is not a retrieval failure in any sense a user would recognise.

4.6 Sensitivity to chunk size

Chunk size	Chunks	Fixed	Broken	Stage-1 misses
450	219	2	0	3
550	167	4	2	2
700	126	1	1	2

At 700 characters the measured benefit nearly disappears: one rescue instead of four. The reason is mechanical. Larger chunks are more likely to contain both a general rule and the exception qualifying it, so the baseline retrieves a passage that happens to include the answer and stops failing – there is less for the re-ranker to repair. At 450 characters the opposite happens: stage-1 misses rise to three, because rules fragment across chunk boundaries and no single chunk holds enough context to score highly.

The consequence is that the benefit reported here is partly a function of a chunking parameter chosen by the author. A different reasonable choice would have produced a visibly different conclusion. Re-ranking is most valuable in a particular regime – chunks small enough to isolate the answer, and a corpus where many chunks are topically similar. Both hold at 550 characters on this document; neither is universal. A report presenting only the 550 row would be overstating a general claim on the strength of a tuned one.

5 Latency Profiling

	Baseline	With re-ranking	Overhead
Mean over 10 queries	8.2 ms	169.8 ms	20.7×
Range	6.8–10.2 ms	144.2–216.5 ms	15.6–27.1×

Measured on CPU. Stage-1 cost is one query embedding (7.8 ms) plus a 167×384 matmul. The cross-encoder costs 14.6 ms per (query, chunk) pair.

An overhead of 20.7× sounds severe and is almost irrelevant in context. The absolute figures are 8.2 ms and 169.8 ms, and both feed an LLM generation step taking one to three seconds. Re-ranking adds roughly 160 ms to a pipeline whose latency is dominated by a cost an order of magnitude larger. The multiplier alarms only when read without units: it compares a very fast operation to a slightly less fast one. Where re-ranking does become expensive is throughput under concurrent load, since the cross-encoder occupies the CPU for 160 ms of real compute while the matmul does not.

6 Scaling: Why Not Cross-Encode Everything?

At a measured 14.6 ms per pair:

Scope	Time per query
Top-10 re-rank (what we do)	0.15 s
This corpus, all 167 chunks	2.44 s
A 100k-chunk corpus	1,458.66 s (24 min)
A 1M-chunk corpus	14,586.64 s (4.1 hours)

For the bi-encoder, chunk vectors depend only on chunks. They are computed once when the index is built and reused unchanged by every subsequent query, so per-query work is one embedding plus a similarity search – which an ANN index makes sublinear in corpus size. Adding a million chunks increases build time and memory but barely changes query latency.

For the cross-encoder the score is a function of the pair. Nothing about a chunk's relevance can be known before the query exists, so no work amortises across queries and cost is strictly linear in candidates scored. The measured 14.6 ms per pair makes this concrete: scoring this 167-chunk corpus exhaustively takes 2.44 seconds per query, and a 100k-chunk corpus would take 24 minutes. Not merely slow – the wrong shape of computation to apply at corpus scale.

Two-stage retrieval resolves the asymmetry by letting each model do what it is cheap at. The bi-encoder reduces the corpus to 10 candidates at index-time cost; the cross-encoder adjudicates those 10 in 0.15 seconds, approximating cross-encoder precision at close to bi-encoder cost.

The arrangement has a hard limit, which this data shows plainly. The re-ranker can only reorder what stage 1 retrieved. Two queries failed in both pipelines because the gold chunk never entered the top 10 – the retaliate and heal-cap queries, both vocabulary mismatches sharing almost no content words with the governing rule. No improvement to the re-ranker could have recovered them. Stage-1 recall is the ceiling; stage 2 only determines how much of it is realised.

7 Conclusion

Three of ten is a real but modest effect, and two qualifications bound it. The chunk-size ablation shows the benefit is largely specific to a size that isolates rules from their exceptions; at 700 characters it nearly vanishes. And the regressions are not noise – they stem from the same preference for elaboration that produces the rescues, which makes re-ranking a trade rather than a free improvement, its net value depending on how often a corpus states answers tersely.

What the results show most clearly is the compression of bi-encoder scores. Ten candidates within a 0.07 band, the correct chunk at rank 7 and 0.066 below a distractor, is not a near miss but an indication that cosine similarity over pooled embeddings had no resolution left to make the distinction. The cross-encoder separated the same candidates across several points. This is the mechanism behind every rescue observed, and the concrete form of the architectural difference described in Section 2.

Given more time the clearest improvement would target stage 1 rather than stage 2. Both unreachable failures were vocabulary mismatches, precisely what a lexical signal handles well; a hybrid BM25 and dense stage would likely have placed both gold chunks in the top 10 and handed them to a re-ranker already shown capable of promoting them. Raising recall at stage 1 lifts the ceiling, whereas further work on stage 2 only redistributes what lies beneath it.

A Reproducing

```
curl -sL https://www.orderofgamers.com/downloads/Gloomhaven_v1.1.pdf \
     -o gloomhaven_rulebook.pdf
export GEMINI_API_KEY="..."
jupyter notebook rag_rerank.ipynb
```

Code: `https://github.com/Sampath7099/RAG-assignment`